UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

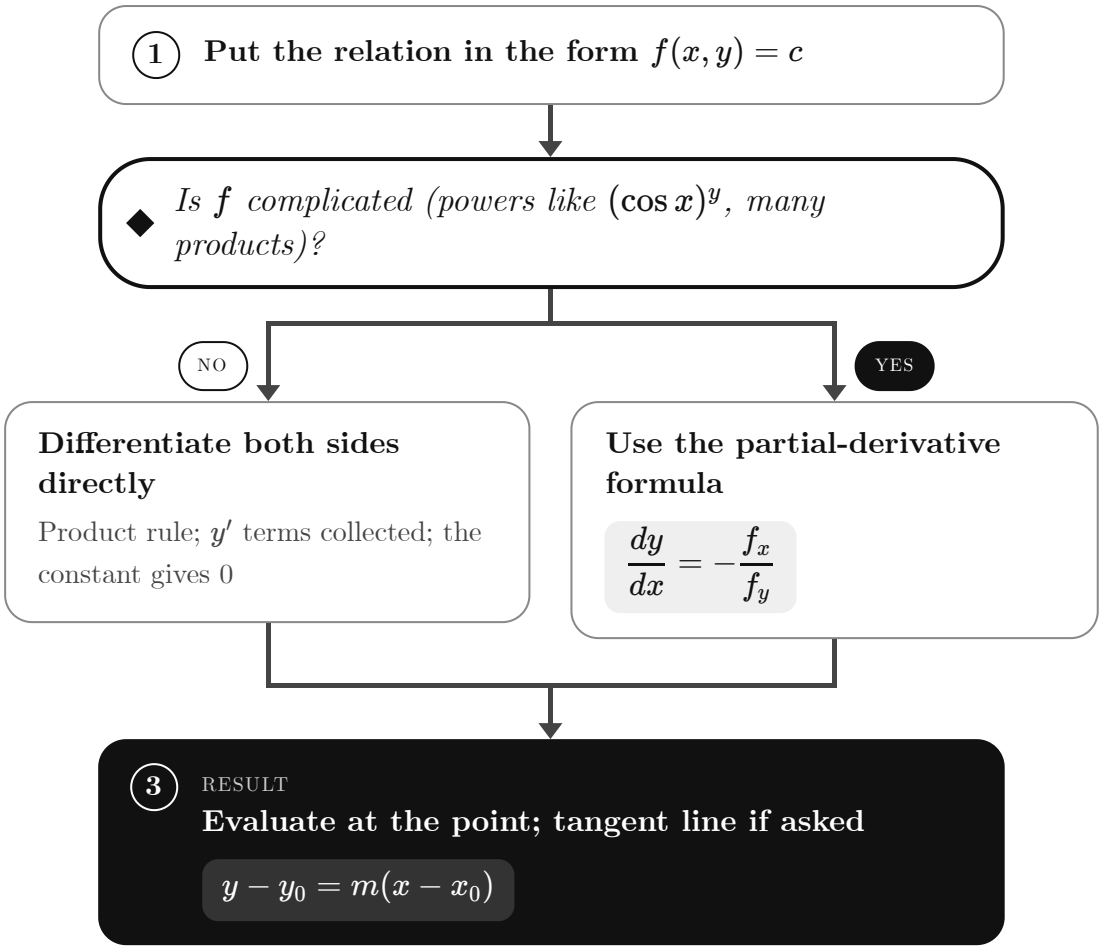
Class 004 · Implicit functions, problems on implicit and composite functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9 · S003·26	Deck 004 / 003, slide number (rendered .pptx)
T004a 00:13:09	Transcript 004 part I: "Implicit functions and problems"
T004b 00:02:31	Transcript 004 part II: "Problems on composite and implicit functions"
T003b 00:10:33	Transcript 003 part II (implicit problems at its end)
CW U1 P17 · ESA Jan-24 s3 1d	Unit 1 class-work problem 17 · past-paper question

Contents

1. Figure F2.1 · Finding dy/dx for an implicit function: two routes.
2. Figure F3.1 · Partial derivatives from $f(x, y, z) = 0$.
3. Figure F4.1 · Which rule? Read the chain first.
4. Figure F4.2 · Proving $u_x + u_y + u_z = 0$ for $u = f(x - y,\, y - z,\, z - x)$.
5. Figure F5.1 · Solving a polar-coordinate composite problem.
6. Total derivative vs chain rule vs implicit differentiation

2 Implicit differentiation in two variables

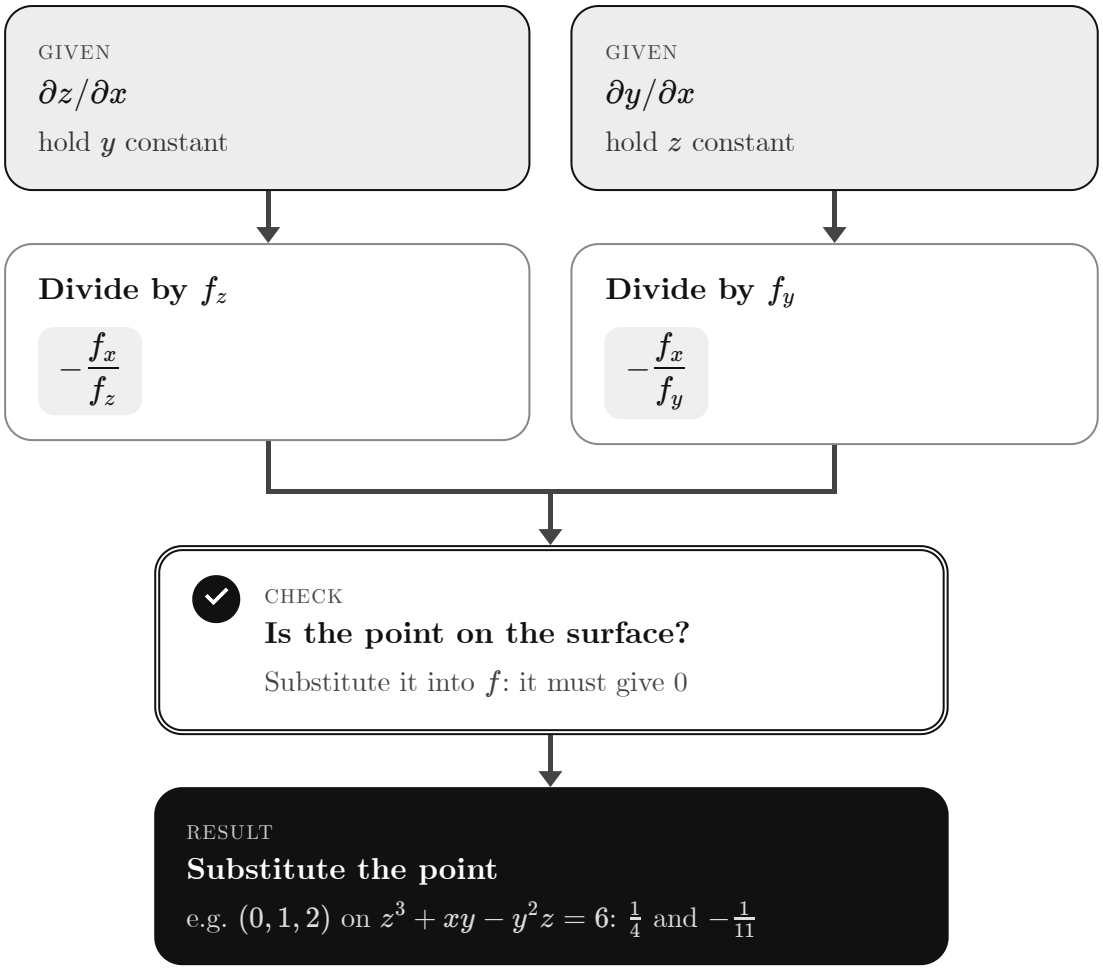


Remark. Both routes give the same dy/dx (W2.1, W2.2). The formula needs $f_y \neq 0$ at the point.

Figure F2.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

3 Implicit functions of three variables

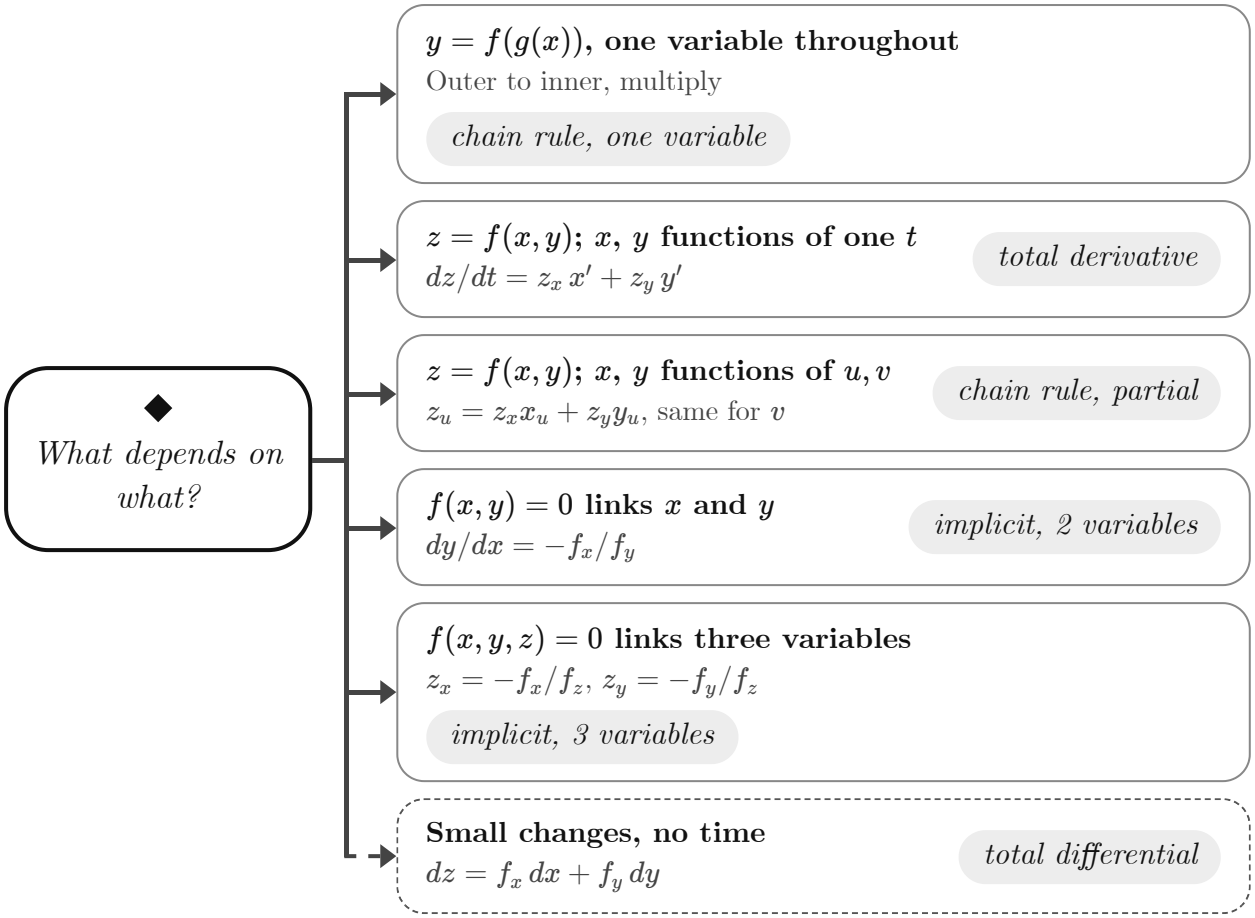


Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

Figure F3.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26

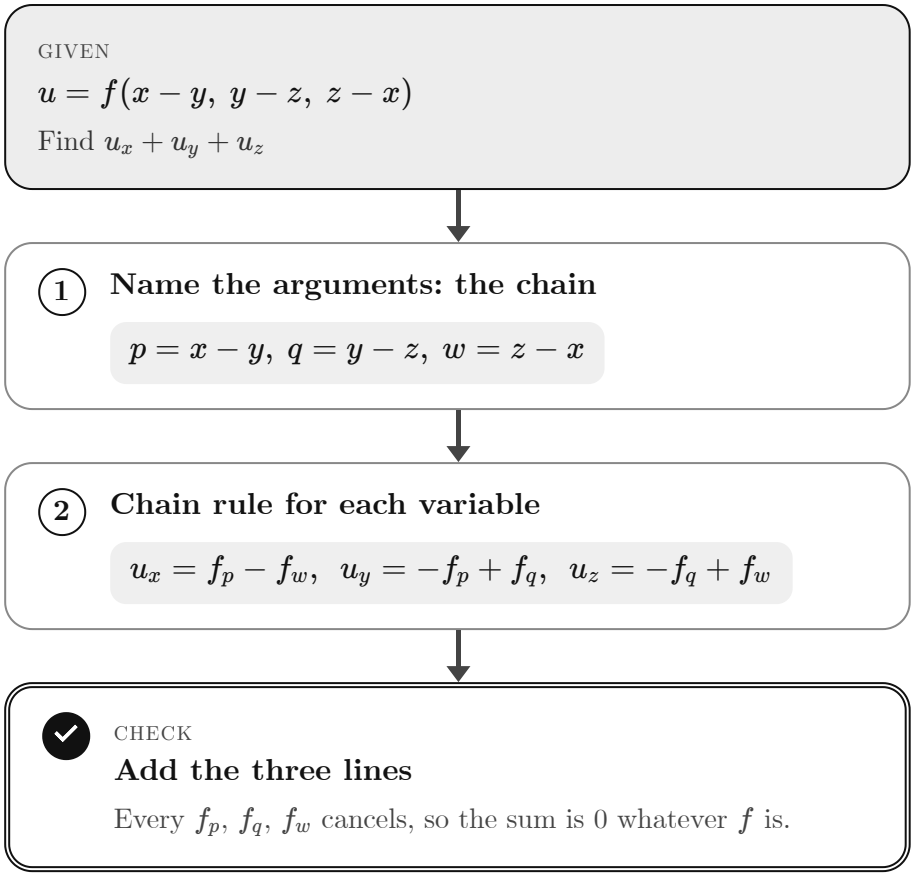
4 Composite functions and the chain rule



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F4.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Remark. Same method for $f(x/y, y/z, z/x)$ with weights x, y, z (W4.3): name the arguments, write each partial, then combine.

Figure F4.2. Proving $u_x + u_y + u_z = 0$ for $u = f(x-y,\ y-z,\ z-x)$. ESA ★★★

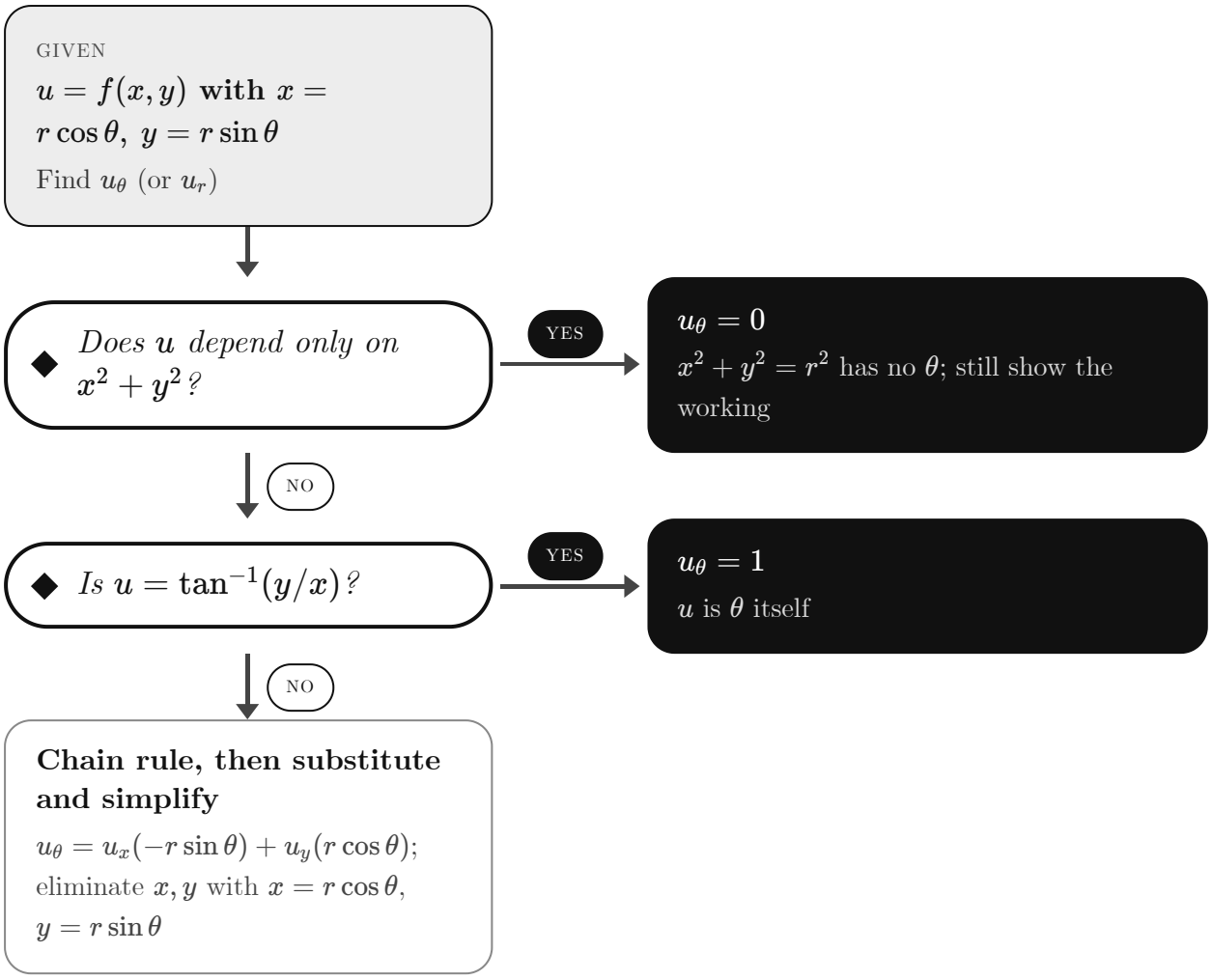
ESA Jan-24 s2 1d · T004b 00:02:31

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x,y)$, $x(t)$, $y(t)$	$z = f(x,y)$, $x(u,v)$, $y(u,v)$	$f(x,y) = 0$ or $f(x,y,z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

5 Composite functions in polar coordinates



Remark. The first two rungs let you predict the answer; the exam still expects the full chain-rule working (W5.1, W5.3, W5.4).

Figure F5.1. Solving a polar-coordinate composite problem. ESA ★★

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